Measure Theory Basics

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The fundamental goal of measure theory is to extend our intuitive notions of length, area, and volume to more abstract and complex sets. This generalization is crucial in many areas of mathematics, including probability theory and functional analysis, which form the backbone of many machine learning and optimization techniques.

1 Basic Concepts

Given a set X, we aim to extend the notions of length, area, and volume to a broader class of sets. This requires identifying "good" sets that can be measured, and understanding the structure that allows us to define a measure.

- The powerset of X, denoted by $P(X)$, is the collection of all subsets of X.
- Our goal is to generalize the concept of length or volume to "good" sets, which do not necessarily include all subsets of X.

1.1 Definition: σ-Algebra

A collection $F \subseteq P(X)$ is called a σ-algebra if it satisfies the following properties:

1. $\emptyset \in F$ and $X \in F$,
2. If $A \in F$, then $A^C \in F$ (closure under complementation),
3. If $\{A_i\}_{i\in\mathbb{N}} \subseteq F$, then $\bigcup_{i=1}^{\infty} A_i \in F$ (closure under countable unions).

A set $A \in F$ is called **F-measurable**.

- The elements of F are the "good" sets that can be measured.
- The smallest σ-algebra is $\{\emptyset, X\}$.
- The largest possible σ-algebra is $P(X)$, but for some sets, such as $X = \mathbb{R}$, $P(X)$ is too large to be a σ-algebra.
- The intersection of any collection of σ-algebras is also a σ-algebra.

1.2 Generated σ-Algebra

Definition: Let $M \subseteq P(X)$. The smallest σ-algebra that contains M is denoted by $\sigma(M)$ and is called the **σ-algebra generated by** M. Formally,

$$\sigma(M) = \bigcap \{F : M \subseteq F \text{ and } F \text{ is a } \sigma\text{-algebra}\}.$$

- Typically, M consists of sets of the form $Y^{-1}(A)$, where $Y : X \to \mathbb{R}$ is a random variable and $A \in F$.

Example: Let $X = \{a, b, c, d\}$ and $M = \{\{a\}, \{b\}\}$. Then,

$$\sigma(M) = \{\emptyset, X, \{a\}, \{b\}, \{b, c, d\}, \{a, c, d\}, \{a, b\}, \{c, d\}\},$$

which has 8 elements, as opposed to $2^4 = 16$ elements in $P(X)$.

1.3 Borel σ-Algebra on X

The σ-algebra generated by the open sets of X is called the Borel σ-algebra and is denoted by $B(X)$.

- X can be a subset of $\mathbb{R}$ or $\mathbb{R}^n$.
- More generally, for a topological space (X, T) or a metric space (X, d), the Borel σ-algebra is $B(X) = \sigma(T)$, where T denotes the collection of open sets in X.

Here, $B(X)$ contains all sets that we can measure.

2 Measure

Let (X, F) be a measurable space. A map $\mu : F \to [0, \infty]$ is called a measure if it satisfies the following properties:

- $\mu(\emptyset) = 0$,
- $\mu(A) \geq 0$ for all $A \in F$,
- $\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$ if $A_i \cap A_j = \emptyset$ for $i \neq j$. This property is called σ-additivity.

The triple (X, F, μ) is called a measure space.

Examples: Consider X with $F = P(X)$, the power set of X.

a) **Counting measure:** For a set $A \subseteq X$,

$$\mu(A) = \begin{cases} |A|, & \text{if } A \text{ is finite,} \\ \infty, & \text{otherwise.} \end{cases}$$

b) **Dirac measure at** $p \in X$: For $A \subseteq X$,

$$\delta_p(A) = \begin{cases} 1, & \text{if } p \in A, \\ 0, & \text{otherwise.} \end{cases}$$

c) **Lebesgue measure on** $[0, 1]^n$: This measure generalizes the notion of volume. We require that $\mu([0, 1]^n) = 1$ and $\mu(x + A) = \mu(A)$. The Lebesgue measure is defined on the Borel σ-algebra of $[0, 1]^n$ (or $\mathbb{R}^n$).

Special Case on $\mathbb{R}$: The measure problem on $P(\mathbb{R})$ with $\mu([a, b]) = b - a$ and $\mu(x + A) = \mu(A)$ does not have a solution, as $P(\mathbb{R})$ is too large.

3 Theorem on Zero Measure

If $\mu([0, 1]) < \infty$ on $P(\mathbb{R})$, then $\mu = 0$.

Proof Outline: The proof relies on defining an equivalence relation $x \sim y$ if $x - y \in \mathbb{Q}$, and using properties of translation-invariance and σ-finiteness.

4 Measurable Maps (or Random Variables)

Let (Ω_1, F_1) and (Ω_2, F_2) be measurable spaces. A map $f : \Omega_1 \to \Omega_2$ is said to be F_2-measurable if for every $A \in F_2$, the pre-image $f^{-1}(A) \in F_1$.

Motivation: In the context of integration, the pre-image $f^{-1}(L)$ must be measurable in order to define the integral over the domain.

Examples:

a) Let (Ω, F) and $(\mathbb{R}, B(\mathbb{R}))$. Define the indicator function $\mathbb{1}_A : \Omega \to \mathbb{R}$ by

$$\mathbb{1}_A(w) = \begin{cases} 1, & \text{if } w \in A, \\ 0, & \text{otherwise.} \end{cases}$$

b) If f is F_1-measurable and g is F_2-measurable, then the composition $g \circ f$ is F_1-measurable.

c) If f and g are F-measurable, then $f + g$, $f - g$, and $|f|$ are also F-measurable.

5 Lebesgue Integral

Consider a simple (step) function:

$$f(x) := \sum_{i=1}^{n} C_i \Pi_{A_i}(x),$$

where $\Pi_{A_i}(x)$ is the indicator function for the set A_i. This function is measurable.

Integral:

$$I(f) := \sum_{i=1}^{n} C_i \mu(A_i),$$

where $\mu(A_i)$ is the measure of the set A_i and $\mu(A_i) = I(\Pi_{A_i})$.
We impose $C_i \geq 0$ to avoid issues with terms like $C_i \infty - C_j \infty$.

Definition: Let S^+ denote the set of non-negative simple functions:

$$S^+ := \{f : X \to \mathbb{R} \mid f \text{ is simple and } f \geq 0\}.$$

Note: Simple functions do not have unique representations.

Lebesgue Integral w.r.t μ: For $f \in S^+$,

$$\int_X f \, d\mu := \sum_{i=1}^{n} C_i \mu(A_i).$$

Here, f is represented as $f = \sum_{i=1}^{n} C_i \Pi_{A_i}$, where Π_{A_i} is the indicator function and $\mu(A_i)$ is the measure of the set A_i.

Properties:

a) Linearity: $\int_X (\alpha f + \beta g)\, d\mu = \alpha \int_X f\, d\mu + \beta \int_X g\, d\mu \quad$ for $\alpha, \beta \geq 0$

b) Monotonicity: If $f \leq g$, then $\int_X f\, d\mu \leq \int_X g\, d\mu \quad$ for $f, g \in S^+$

Definition: Let $f : X \to \mathbb{R}^+$ be F-measurable. The Lebesgue integral of f is defined by

$$\int_X f\, d\mu := \sup \left\{ \int_X h\, d\mu \mid h \in S^+,\ h \leq f \right\}.$$

The function f is said to be μ-integrable if $\int_X f\, d\mu < \infty$.

Side definition: We say $f = g$ μ-almost everywhere (a.e.) if

$$\mu\left(\{x \in X \mid f(x) \neq g(x)\}\right) = 0.$$

Properties of the Lebesgue Integral:

a) If $f = g$ μ-a.e., then $\int_X f\, d\mu = \int_X g\, d\mu$.

b) If $f = 0$ μ-a.e., then $\int_X f\, d\mu = 0$.

c) If $f \leq g$ μ-a.e., then $\int_X f\, d\mu \leq \int_X g\, d\mu$.

6 Monotone Convergence Theorem

Let (X, F, μ) be a measure space, and let $\{f_n\}$ be a sequence of F-measurable functions $f_n : X \to [0, \infty)$ such that

$$f_1 \leq f_2 \leq \cdots \leq f_n \ \mu\text{-a.e.}$$

and

$$\lim_{n\to\infty} f_n(x) = f(x) \ \mu\text{-a.e.}.$$

Then

$$\lim_{n\to\infty} \int_X f_n\, d\mu = \int_X f\, d\mu.$$

Application: If $\{g_n\}$ is a sequence of non-negative measurable functions, then

$$\int_X \sum_{n=1}^{\infty} g_n\, d\mu = \sum_{n=1}^{\infty} \int_X g_n\, d\mu.$$

7 Measures on Product Spaces

Let $(X, \mathcal{F}_X, \mu_X)$ and $(Y, \mathcal{F}_Y, \mu_Y)$ be measure spaces. The product σ-algebra $\mathcal{F}_X \otimes \mathcal{F}_Y$ is the smallest σ-algebra on $X \times Y$ that contains all rectangles of the form $A \times B$, where $A \in \mathcal{F}_X$ and $B \in \mathcal{F}_Y$.

Definition: A measure μ on the product space $X \times Y$ is called a product measure if it is the unique measure such that for all rectangles $A \times B$,

$$\mu(A \times B) = \mu_X(A) \cdot \mu_Y(B).$$

Theorem (Fubini's Theorem): If $(X, \mathcal{F}_X, \mu_X)$ and $(Y, \mathcal{F}_Y, \mu_Y)$ are σ-finite measure spaces, and $f : X \times Y \to \mathbb{R}$ is a measurable function, then

$$\int_{X\times Y} f\, d(\mu_X \otimes \mu_Y) = \int_X \left(\int_Y f(x, y)\, d\mu_Y(y) \right) d\mu_X(x).$$

8 Radon Measures

Definition: A measure μ on a topological space X is called a Radon measure if it is locally finite and inner regular. This means that for every Borel set $B \subseteq X$,

$$\mu(B) = \sup\{\mu(K) \mid K \subseteq B,\ K \text{ compact}\}.$$

Theorem (Riesz Representation Theorem): Let X be a locally compact Hausdorff space. Every continuous linear functional on the space of continuous functions with compact support can be represented as integration with respect to a Radon measure.

9 Convergence of Measures

Definition: A sequence of measures $\{\mu_n\}$ on a measurable space $(X, \mathcal{F})$ converges weakly to a measure μ if for every bounded continuous function f,

$$\lim_{n\to\infty} \int_X f \, d\mu_n = \int_X f \, d\mu.$$

Theorem (Portmanteau Theorem): Let $\{\mu_n\}$ be a sequence of probability measures on $(X, \mathcal{F})$. The following are equivalent:

1. $\mu_n \xrightarrow{w} \mu$.
2. $\liminf_{n\to\infty} \mu_n(A) \geq \mu(A)$ for every closed set A.
3. $\limsup_{n\to\infty} \mu_n(B) \leq \mu(B)$ for every open set B.

10 Cauchy Sequences

Definition: A sequence $\{x_n\}$ in a metric space (X, d) is called a **Cauchy sequence** if for every $\epsilon > 0$, there exists a positive integer N such that for all $m, n \geq N$,

$$d(x_n, x_m) < \epsilon.$$

In other words, the elements of the sequence become arbitrarily close to each other as n and m get large.

Example 1: Consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{n}$ in the metric space $(\mathbb{R}, d)$, where $d(x, y) = |x - y|$. This sequence is a Cauchy sequence because for any $\epsilon > 0$, we can find N such that for all $m, n \geq N$,

$$|x_n - x_m| = \left|\frac{1}{n} - \frac{1}{m}\right| \leq \frac{1}{N}.$$

Since $\frac{1}{N}$ can be made arbitrarily small by choosing N large enough, $\{x_n\}$ is a Cauchy sequence.

Example 2: Consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{n^2}$ in the metric space $(\mathbb{R}, d)$, where $d(x, y) = |x - y|$. This sequence is also a Cauchy sequence because for any $\epsilon > 0$, we can find N such that for all $m, n \geq N$,

$$|x_n - x_m| = \left|\frac{1}{n^2} - \frac{1}{m^2}\right| \leq \frac{1}{N^2}.$$

Again, $\frac{1}{N^2}$ can be made arbitrarily small by choosing N large enough, so $\{x_n\}$ is a Cauchy sequence.

11 Complete Inner Product Space

Definition: A **complete inner product space** is an inner product space $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ that is also a complete metric space with respect to the metric induced by the inner product. Specifically, $\mathcal{H}$ is a complete inner product space if every Cauchy sequence $\{x_n\}$ in $\mathcal{H}$ converges to an element in $\mathcal{H}$.

In other words, $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ is complete if:

$$\text{If } \{x_n\} \text{ is a Cauchy sequence in } \mathcal{H}, \text{ then } \exists x \in \mathcal{H} \text{ such that } \lim_{n \to \infty} x_n = x.$$

11.1 Examples

Example 1: The space of square-integrable functions $L^2([a, b])$ is a complete inner product space. Here, the inner product is defined by:

$$\langle f, g \rangle = \int_a^b f(x) g(x) \, dx,$$

and the space $L^2([a, b])$ consists of all functions f such that:

$$\int_a^b |f(x)|^2 \, dx < \infty.$$

The space $L^2([a, b])$ is complete with respect to the norm:

$$\|f\| = \left(\int_a^b |f(x)|^2 \, dx \right)^{1/2}.$$

Example 2: The Euclidean space $\mathbb{R}^n$ with the standard inner product is a complete inner product space. In this case, the inner product is given by:

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i,$$

where $x = (x_1, x_2, \ldots, x_n)$ and $y = (y_1, y_2, \ldots, y_n)$. The space $\mathbb{R}^n$ is complete with respect to the Euclidean norm:

$$\|x\| = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}.$$

12 Hilbert Space

Definition: A **Hilbert space** $\mathcal{H}$ is a complete inner product space. This means that $\mathcal{H}$ is a vector space equipped with an inner product $\langle \cdot, \cdot \rangle$ such that $\mathcal{H}$ is complete with respect to the norm induced by the inner product:

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Specifically, $\mathcal{H}$ is complete if every Cauchy sequence in $\mathcal{H}$ converges to a limit in $\mathcal{H}$.

Examples:

1. **Euclidean Space:** Consider $\mathbb{R}^n$ with the standard inner product
$$\langle x, y \rangle = \sum_{i=1}^{n} x_i y_i.$$
The norm is given by
$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{i=1}^{n} x_i^2}.$$
$\mathbb{R}^n$ with this inner product is a finite-dimensional Hilbert space, as it is complete with respect to this norm.

2. **Space of Square Integrable Functions:** Let $\mathcal{L}^2([a, b])$ be the space of all functions f defined on $[a, b]$ for which
$$\int_a^b |f(x)|^2 \, dx < \infty.$$
The inner product is defined by
$$\langle f, g \rangle = \int_a^b f(x)\overline{g(x)} \, dx.$$
The norm is
$$\|f\| = \sqrt{\langle f, f \rangle} = \left(\int_a^b |f(x)|^2 \, dx \right)^{1/2}.$$
$\mathcal{L}^2([a, b])$ is a Hilbert space because it is complete with respect to this norm, meaning every Cauchy sequence of functions in $\mathcal{L}^2([a, b])$ converges to a function in $\mathcal{L}^2([a, b])$.

13 Measures and Integration in Hilbert Spaces

Definition: Let $\mathcal{H}$ be a Hilbert space. A measure μ on $\mathcal{H}$ is called a Radon measure if for every bounded linear functional L on $\mathcal{H}$, the map $L \mapsto \int_{\mathcal{H}} L \, d\mu$ is continuous.

Theorem (Hilbert Space Integration): If $f : \mathcal{H} \to \mathbb{R}$ is a measurable function and μ is a Radon measure on $\mathcal{H}$, then
$$\int_{\mathcal{H}} f \, d\mu$$
is well-defined and the integral is taken with respect to the inner product structure of $\mathcal{H}$.

14 Empirical Measures

Definition: Given a sample $\{x_1, x_2, \ldots, x_n\} \subseteq X$, the empirical measure $\hat{\mu}_n$ is defined by
$$\hat{\mu}_n(A) := \frac{1}{n} \sum_{i=1}^{n} \delta_{x_i}(A),$$
where δ_{x_i} is the Dirac measure at x_i.

Theorem (Weak Convergence of Empirical Measures): If $\{x_i\}$ are i.i.d. random variables with distribution μ, then
$$\hat{\mu}_n \xrightarrow{w} \mu$$
almost surely as $n \to \infty$.